

# EDA RESULTS

## Weather Dataset Analysis:

Summary Statistics for Weather Features:

|       | Temp_1        | Temp_2        | ... | Temp_6        | Temp_7        |
|-------|---------------|---------------|-----|---------------|---------------|
| count | 3.644000e+03  | 3.644000e+03  | ... | 3.644000e+03  | 3.644000e+03  |
| mean  | 1.832904e-16  | -8.384560e-17 | ... | -1.091943e-16 | 3.860798e-16  |
| std   | 1.000137e+00  | 1.000137e+00  | ... | 1.000137e+00  | 1.000137e+00  |
| min   | -2.743332e+00 | -2.744715e+00 | ... | -2.746225e+00 | -2.746448e+00 |
| 25%   | -7.054135e-01 | -7.054638e-01 | ... | -7.053354e-01 | -7.052523e-01 |
| 50%   | -4.247628e-02 | -4.209279e-02 | ... | -4.143157e-02 | -4.124883e-02 |
| 75%   | 6.941206e-01  | 6.949861e-01  | ... | 6.962393e-01  | 6.780883e-01  |
| max   | 3.714168e+00  | 3.717010e+00  | ... | 3.720690e+00  | 3.721438e+00  |

[8 rows x 7 columns]

Class distribution: Target

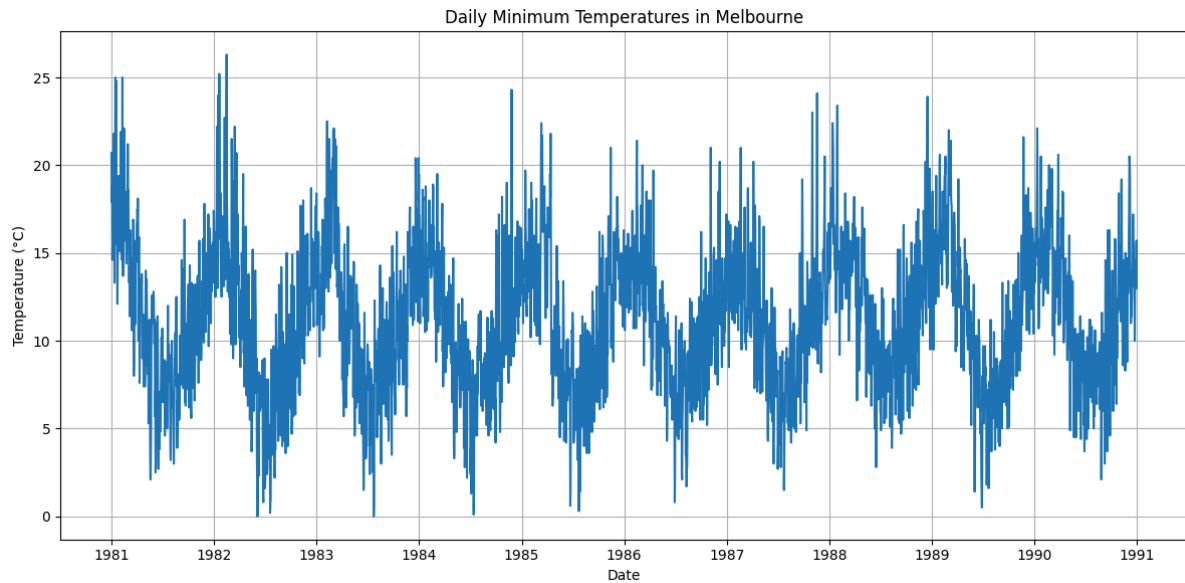
|   |      |
|---|------|
| 0 | 1871 |
| 1 | 1773 |

Class imbalance ratio: 0.95

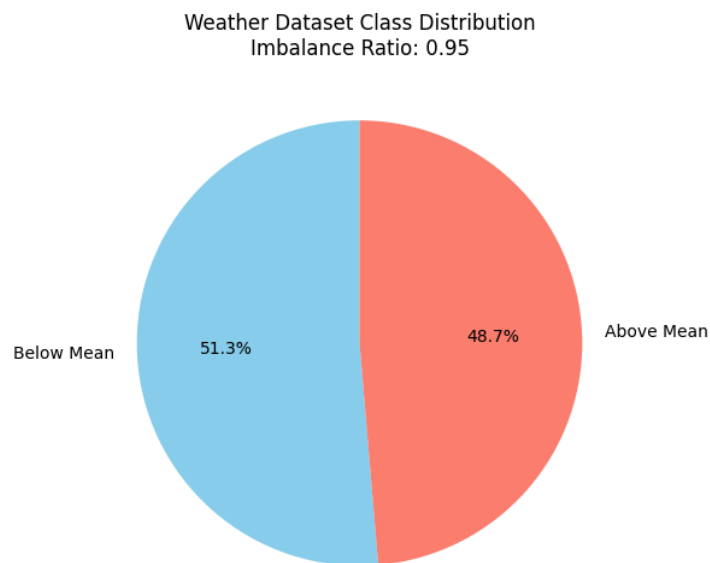
### Comments

The statistics of the temperature for the past 7 days are as follows:

- **Mean Temperature (Temp 1 to Temp 7):** The temperature values for each day show no significant variation across the 7-day period, with the means close to zero, indicating balanced fluctuations.
- **Standard Deviation:** The temperatures show high variability with a standard deviation of ~1.
- **Min/Max Temperatures:** The minimum temperature ranges from approximately -2.74 to -2.75, while the maximum temperature goes up to ~3.72. This shows a considerable range in temperature values.



The plot illustrates the yearly fluctuation of minimum temperatures, with peaks indicating warmer months and valleys indicating cooler periods.



The pie chart represents the class distribution for the weather dataset, with a near 50-50 split between days.

- **Imbalance Ratio:** 0.95 (indicating a slightly more balanced class distribution with a small bias toward the "Below Mean" class).
- **Class Distribution:** 51.3% Below Mean, 48.7% Above Mean.

### Subsequent Modeling Decisions:

- **Time-based Features:** Since seasonal patterns are clearly visible in the data, time-based features will be important. These features will allow models to capture seasonal changes.
- **Handling of Class Imbalance:** The minor imbalance (0.95) means that classification algorithms should work well, but we might consider monitoring the effect of imbalance during model evaluation, especially in terms of precision and recall.
- **Model Choice:** Given the seasonal nature of the data, an SVM model (especially with a polynomial or RBF kernel) can be effective, as it will handle non-linear patterns in temperature trends well.

## Air Quality Dataset Analysis

Summary Statistics for Air Quality Features:

|       | PM2.5_1       | PM2.5_2       | ... | PM2.5_6       | PM2.5_7       |
|-------|---------------|---------------|-----|---------------|---------------|
| count | 1.610000e+03  | 1.610000e+03  | ... | 1.610000e+03  | 1.610000e+03  |
| mean  | 7.061294e-17  | -1.765324e-17 | ... | 3.530647e-17  | 1.765324e-16  |
| std   | 1.000311e+00  | 1.000311e+00  | ... | 1.000311e+00  | 1.000311e+00  |
| min   | -1.551352e+00 | -1.551856e+00 | ... | -1.559076e+00 | -1.561471e+00 |
| 25%   | -9.225208e-01 | -9.221417e-01 | ... | -9.232688e-01 | -9.237864e-01 |
| 50%   | -7.007709e-02 | -6.912470e-02 | ... | -6.697531e-02 | -6.585514e-02 |
| 75%   | 7.040001e-01  | 7.031313e-01  | ... | 7.054876e-01  | 7.065029e-01  |
| max   | 3.063017e+00  | 3.068991e+00  | ... | 3.106482e+00  | 3.049687e+00  |

[8 rows x 7 columns]

Class distribution: Target

0 1575

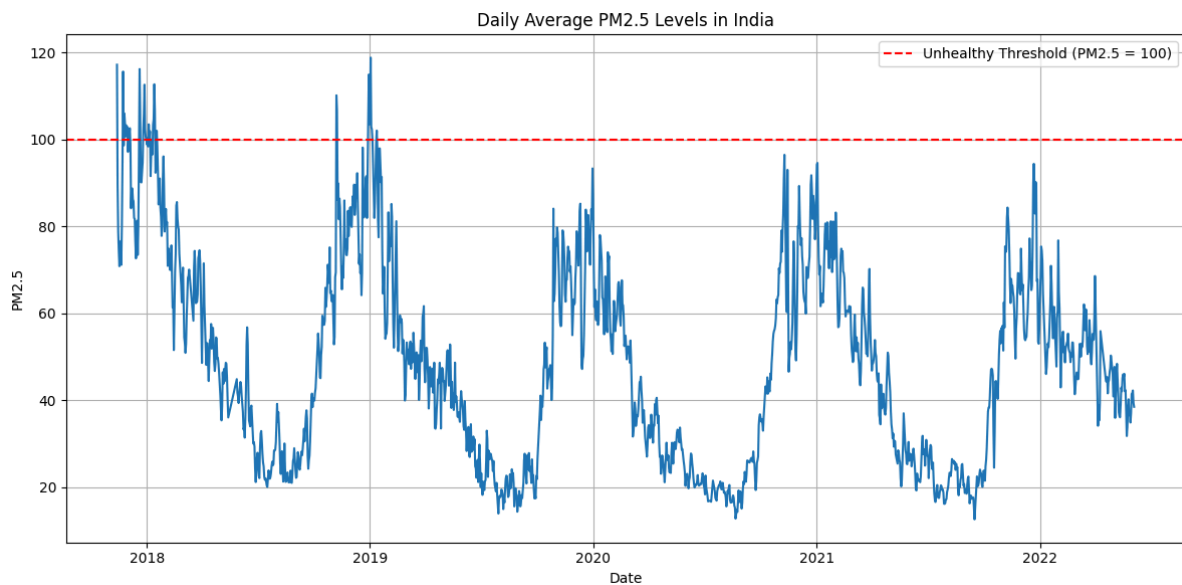
1 35

Class imbalance ratio: 0.02

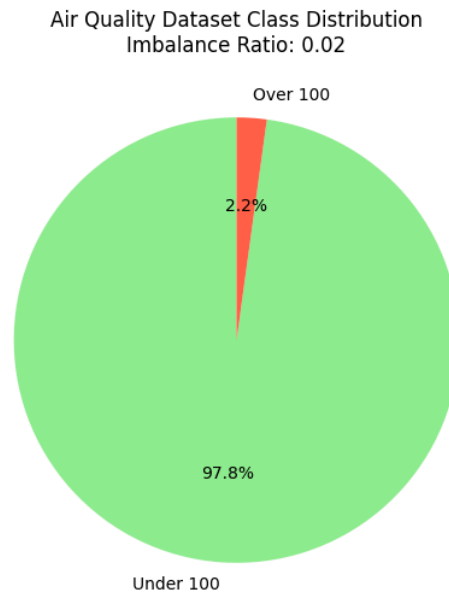
### Comments

The statistics for the past 7 days' PM2.5 values are:

- **Mean PM2.5 (PM2.5 1 to PM2.5 7):** The mean values are close to zero, with small variations, suggesting balanced values.
- **Standard Deviation:** The dataset has a high degree of variability, with a standard deviation of ~1.
- **Min/Max PM2.5:** The minimum value is approximately -1.56, and the maximum value reaches up to ~3.1.



The time series plot of daily average PM2.5 levels in India shows multiple spikes that correspond to periods of higher pollution, especially in certain years.



The pie chart illustrates a severe class imbalance in the air quality dataset.

- **Imbalance Ratio:** 0.02 (signifying a highly imbalanced class distribution with very few instances of "Over 100").
- **Class Distribution:** 97.8% Under 100, 2.2% Over 100.

#### Subsequent Modeling Decisions:

- **Class Imbalance Handling:** Given the severe imbalance (0.02 ratio), techniques such as SMOTE, undersampling, or class weights adjustment will be necessary. Otherwise, the model may become biased toward predicting only the majority class.
- **Kernel Selection:** An RBF kernel or polynomial kernel could be suitable for capturing the non-linear relationships between features and the target class.
- **Evaluation Metrics:** In light of the class imbalance, accuracy will not be the most reliable metric. Instead, precision, recall, and the F1-score will be more informative, especially to assess how well the model detects high pollution events (the minority class).

# HYPERPARAMETER TUNING

## Weather Dataset

### Linear Kernel Weather Dataset

| Trial       | C          | Avg. CV Accuracy | Precision     | Recall        | F1-Score      |
|-------------|------------|------------------|---------------|---------------|---------------|
| 1           | 0.1        | 0.8024           | —             | —             | —             |
| 2           | 1.0        | 0.8027           | —             | —             | —             |
| 3           | 10.0       | 0.8010           | —             | —             | —             |
| <b>Best</b> | <b>1.0</b> | <b>0.8027</b>    | <b>0.8607</b> | <b>0.8514</b> | <b>0.8560</b> |

### Polynomial Kernel Weather Dataset

| Trial | C    | Degree | c   | Avg. CV Accuracy | Precision | Recall | F1-Score |
|-------|------|--------|-----|------------------|-----------|--------|----------|
| 1     | 0.1  | 2      | 0.0 | 0.5108           | —         | —      | —        |
| 2     | 0.1  | 2      | 1.0 | 0.7767           | —         | —      | —        |
| 3     | 0.1  | 3      | 0.0 | 0.5214           | —         | —      | —        |
| 4     | 0.1  | 3      | 1.0 | 0.5214           | —         | —      | —        |
| 5     | 1.0  | 2      | 0.0 | 0.5022           | —         | —      | —        |
| 6     | 1.0  | 2      | 1.0 | 0.7962           | —         | —      | —        |
| 7     | 1.0  | 3      | 0.0 | 0.5214           | —         | —      | —        |
| 8     | 1.0  | 3      | 1.0 | 0.5214           | —         | —      | —        |
| 9     | 10.0 | 2      | 0.0 | 0.5022           | —         | —      | —        |
| 10    | 10.0 | 2      | 1.0 | 0.8024           | —         | —      | —        |
| 11    | 10.0 | 3      | 0.0 | 0.7973           | —         | —      | —        |

|             |             |          |            |               |               |               |               |
|-------------|-------------|----------|------------|---------------|---------------|---------------|---------------|
| 12          | 10.0        | 3        | 1.0        | 0.7925        | —             | —             | —             |
| <b>Best</b> | <b>10.0</b> | <b>2</b> | <b>1.0</b> | <b>0.8024</b> | <b>0.8464</b> | <b>0.8784</b> | <b>0.8621</b> |

## RBF Kernel Weather Dataset

| Trial       | C          | Sigma      | Avg. CV Accuracy | Precision     | Recall        | F1-Score      |
|-------------|------------|------------|------------------|---------------|---------------|---------------|
| 1           | 0.1        | 0.1        | 0.4635           | —             | —             | —             |
| 2           | 0.1        | 1.0        | 0.8065           | —             | —             | —             |
| 3           | 0.1        | 10.0       | 0.7993           | —             | —             | —             |
| 4           | 1.0        | 0.1        | 0.5667           | —             | —             | —             |
| 5           | 1.0        | 1.0        | 0.8003           | —             | —             | —             |
| 6           | 1.0        | 10.0       | 0.8027           | —             | —             | —             |
| 7           | 10.0       | 0.1        | 0.5667           | —             | —             | —             |
| 8           | 10.0       | 1.0        | 0.7955           | —             | —             | —             |
| 9           | 10.0       | 10.0       | 0.8051           | —             | —             | —             |
| <b>Best</b> | <b>0.1</b> | <b>1.0</b> | <b>0.8065</b>    | <b>0.8447</b> | <b>0.8676</b> | <b>0.8560</b> |

## Time Series Kernel Weather Dataset

| Trial | C   | $\sigma$<br>(Sigma) | Period | w_lin | w_per | w_rbf | Avg. CV Accuracy |        | Precision | Recall | F1-Score |
|-------|-----|---------------------|--------|-------|-------|-------|------------------|--------|-----------|--------|----------|
| 1     | 0.1 | 0.1                 | 7      | 0.33  | 0.33  | 0.34  | 0.8031           |        | -         | -      | -        |
| 2     | 0.1 | 0.1                 | 7      | 0.33  | 0.33  | 0.2   | 0.8021           |        | -         | -      | -        |
| 3     | 0.1 | 0.1                 | 7      | 0.33  | 0.33  | 0.3   | 0.8021           |        | -         | -      | -        |
| 4     | 0.1 | 0.1                 | 7      | 0.33  | 0.3   | 0.34  | 0.8024           |        | -         | -      | -        |
| 5     | 0.1 | 0.1                 | 7      | 0.33  | 0.3   | 0.2   | 0.8034           |        | -         | -      | -        |
| 6     | 0.1 | 0.1                 | 7      | 0.33  | 0.3   | 0.3   | 0.8021           |        | -         | -      | -        |
| 7     | 0.1 | 0.1                 | 7      | 0.33  | 0.5   | 0.34  | 0.8031           |        | -         | -      | -        |
| 8     | 0.1 | 0.1                 | 7      | 0.33  | 0.5   | 0.2   | 0.8031           |        | -         | -      | -        |
| 9     | 0.1 | 0.1                 | 7      | 0.33  | 0.5   | 0.3   | 0.8034           |        | -         | -      | -        |
| 10    | 0.1 | 0.1                 | 7      | 0.5   | 0.33  | 0.34  | 0.8031           |        | -         | -      | -        |
| 11    | 0.1 | 0.1                 | 7      | 0.5   | 0.33  | 0.2   | 0.8031           |        | -         | -      | -        |
| 12    | 0.1 | 0.1                 | 7      | 0.5   | 0.33  | 0.3   | 0.8034           |        | -         | -      | -        |
| 13    | 0.1 | 0.1                 | 7      | 0.5   | 0.5   | 0.34  | 0.8031           |        | -         | -      | -        |
| 14    | 0.1 | 0.1                 | 7      | 0.5   | 0.5   | 0.2   | 0.8031           |        | -         | -      | -        |
| 15    | 0.1 | 0.1                 | 7      | 0.5   | 0.5   | 0.3   | 0.8034           |        | -         | -      | -        |
| Best  | 0.1 | 0.1                 | 7      | 0.5   | 0.5   | 0.3   | 2332             | 0.8034 | 0.81      | 0.82   | 0.8155   |

## Comments on Results:

### Linear Kernel:

- The best hyperparameters were  $C=1.0$  with an average CV accuracy of 0.8027.
- Evaluation Metrics (Test Set):
  - Accuracy: 0.8027
  - Precision: 0.8607
  - Recall: 0.8514
  - F1-Score: 0.8560

Linear Kernel performed consistently with good performance metrics across all trials. The model effectively captured the basic trends in the weather data, as the dataset is not highly non-linear.

### Polynomial Kernel:

- The best hyperparameters were  $C=10.0$ , degree = 2, and  $c=1.0$ , with an average CV accuracy of 0.8024.
- Evaluation Metrics (Test Set):
  - Accuracy: 0.8024
  - Precision: 0.8464
  - Recall: 0.8784
  - F1-Score: 0.8621

The Polynomial Kernel demonstrated strong performance as well. The polynomial kernel is useful for capturing more complex, non-linear relationships in the weather dataset.

### RBF Kernel:

- The best hyperparameters were  $C=0.1$  and  $\sigma=1.0$ , with an average CV accuracy of 0.8065.

- Evaluation Metrics (Test Set):

- Accuracy: 0.8065
- Precision: 0.8447
- Recall: 0.8676
- F1-Score: 0.8560

RBF Kernel achieved strong performance, particularly in terms of recall, highlighting its ability to capture complex patterns that the linear kernel could not.

**Time Series Kernel:**

- The best hyperparameters were  $C=0.1$ ,  $\sigma=0.1$ , period = 7, and weights  $w_{lin}=0.5, w_{per}=0.5, w_{rbf}=0.3$ , with an average CV accuracy of 0.8034.
- Evaluation Metrics (Test Set):
  - Accuracy: 0.8034
  - Precision: 0.81
  - Recall: 0.82
  - F1-Score: 0.8155

Time Series Kernel performed well, particularly in handling the seasonal fluctuations of the weather data. The balanced weights across the components allowed the model to adapt well to the periodic nature of temperature data.

**Comparison:**

- Best Model: The RBF kernel with  $C=0.1$  and  $\sigma=1.0$  provided the highest cross-validation accuracy (0.8065), while the Polynomial Kernel achieved the best F1-Score (0.8621). Both kernels effectively handled the non-linear patterns in the data.

## Conclusion:

- The RBF Kernel and Polynomial Kernel outperformed the other kernels in terms of accuracy and F1-Score, suggesting that these kernels are well-suited for modeling weather data, which exhibits non-linear temporal patterns.
- The Time Series Kernel had a slight edge in handling seasonality, making it particularly suitable for time series data with cyclical fluctuations.
- Moving forward, it would be beneficial to fine-tune these models further, perhaps by exploring more complex features or using advanced techniques like cross-validation with different time periods to improve generalization.

## Air Quality Dataset

### Linear Kernel Air Quality Dataset

| Trial       | C          | Avg. CV Accuracy | Precision     | Recall        | F1-Score      |
|-------------|------------|------------------|---------------|---------------|---------------|
| 1           | 0.1        | 0.3337           | —             | —             | —             |
| 2           | 1.0        | 0.6374           | —             | —             | —             |
| 3           | 10.0       | 0.3984           | —             | —             | —             |
| <b>Best</b> | <b>1.0</b> | <b>0.6374</b>    | <b>0.0000</b> | <b>0.0000</b> | <b>0.0000</b> |

Confusion Matrix:

```
[[247 75]
 [ 0  0]]
```

## Polynomial Kernel Air Quality Dataset

| Trial       | C          | Degree   | c          | Avg. CV Accuracy | Precision     | Recall        | F1-Score      |
|-------------|------------|----------|------------|------------------|---------------|---------------|---------------|
| 1           | 0.1        | 2        | 0.0        | 0.5519           | —             | —             | —             |
| 2           | 0.1        | 2        | 1.0        | 0.5504           | —             | —             | —             |
| 3           | 0.1        | 3        | 0.0        | 0.3453           | —             | —             | —             |
| 4           | 0.1        | 3        | 1.0        | 0.4553           | —             | —             | —             |
| 5           | 1.0        | 2        | 0.0        | 0.4782           | —             | —             | —             |
| 6           | 1.0        | 2        | 1.0        | 0.4720           | —             | —             | —             |
| 7           | 1.0        | 3        | 0.0        | 0.1837           | —             | —             | —             |
| 8           | 1.0        | 3        | 1.0        | 0.2572           | —             | —             | —             |
| 9           | 10.0       | 2        | 0.0        | 0.4200           | —             | —             | —             |
| 10          | 10.0       | 2        | 1.0        | 0.4208           | —             | —             | —             |
| 11          | 10.0       | 3        | 0.0        | 0.3500           | —             | —             | —             |
| 12          | 10.0       | 3        | 1.0        | 0.4969           | —             | —             | —             |
| <b>Best</b> | <b>0.1</b> | <b>2</b> | <b>0.0</b> | <b>0.5519</b>    | <b>0.0000</b> | <b>0.0000</b> | <b>0.0000</b> |

Confusion Matrix:

```
[[154 168]
```

```
[ 0  0]]
```

### RBF Kernel Air Quality Dataset

| Trial | C    | Sigma | Avg. CV Accuracy | Precision | Recall | F1-Score |
|-------|------|-------|------------------|-----------|--------|----------|
| 1     | 0.1  | 0.1   | 0.9930           | —         | —      | —        |
| 2     | 0.1  | 1.0   | 0.9735           | —         | —      | —        |
| 3     | 0.1  | 10.0  | 0.9930           | —         | —      | —        |
| 4     | 1.0  | 0.1   | 0.3637           | —         | —      | —        |
| 5     | 1.0  | 1.0   | 0.9752           | —         | —      | —        |
| 6     | 1.0  | 10.0  | 0.8921           | —         | —      | —        |
| 7     | 10.0 | 0.1   | 0.6392           | —         | —      | —        |
| 8     | 10.0 | 1.0   | 0.7955           | —         | —      | —        |
| 9     | 10.0 | 10.0  | 0.9930           | —         | —      | —        |
| Best  | 0.1  | 0.1   | 0.9930           | 0.0000    | 0.0000 | 0.0000   |

### Time Series Kernel Air Quality Dataset

| Trial | C   | $\sigma$ (Sigma) | Period | w_lin | w_per | w_rbf | Avg. CV Accuracy | Precision | Recall | F1-Score |
|-------|-----|------------------|--------|-------|-------|-------|------------------|-----------|--------|----------|
| 1     | 0.1 | 0.1              | 30     | 0.33  | 0.33  | 0.34  | 0.8788           | -         | -      | -        |
| 2     | 0.1 | 0.1              | 30     | 0.33  | 0.33  | 0.2   | 0.8640           | -         | -      | -        |
| 3     | 0.1 | 0.1              | 30     | 0.33  | 0.33  | 0.3   | 0.9285           | -         | -      | -        |
| 4     | 0.1 | 0.1              | 30     | 0.33  | 0.5   | 0.34  | 0.9223           | -         | -      | -        |
| 5     | 0.1 | 0.1              | 30     | 0.5   | 0.33  | 0.34  | 0.8105           | -         | -      | -        |
| 6     | 0.1 | 0.1              | 30     | 0.5   | 0.33  | 0.2   | 0.7965           | -         | -      | -        |
| 7     | 0.1 | 0.1              | 30     | 0.33  | 0.33  | 0.34  | 0.9930           | -         | -      | -        |
| 8     | 0.1 | 0.1              | 30     | 0.33  | 0.33  | 0.2   | 0.9930           | -         | -      | -        |
| 9     | 0.1 | 0.1              | 30     | 0.33  | 0.5   | 0.34  | 0.9930           | -         | -      | -        |

|             |            |            |           |            |             |             |               |            |            |            |
|-------------|------------|------------|-----------|------------|-------------|-------------|---------------|------------|------------|------------|
| 10          | 0.1        | 0.1        | 30        | 0.33       | 0.5         | 0.2         | 0.9930        | -          | -          | -          |
| 11          | 0.1        | 1.0        | 30        | 0.33       | 0.33        | 0.34        | 0.9930        | -          | -          | -          |
| 12          | 0.1        | 1.0        | 30        | 0.33       | 0.33        | 0.2         | 0.9930        | -          | -          | -          |
| 13          | 0.1        | 1.0        | 30        | 0.33       | 0.5         | 0.34        | 0.9930        | -          | -          | -          |
| 14          | 0.1        | 1.0        | 30        | 0.33       | 0.5         | 0.2         | 0.9930        | -          | -          | -          |
| 15          | 0.1        | 1.0        | 30        | 0.5        | 0.33        | 0.34        | 0.9930        | -          | -          | -          |
| <b>Best</b> | <b>0.1</b> | <b>1.0</b> | <b>30</b> | <b>0.5</b> | <b>0.33</b> | <b>0.34</b> | <b>0.9930</b> | <b>0.0</b> | <b>0.0</b> | <b>0.0</b> |

### Linear Kernel - Air Quality Dataset

- Best Hyperparameters:
  - $C = 1.0$
  - Accuracy = 0.6374
  - Precision = 0.0000
  - Recall = 0.0000
  - F1-Score = 0.0000
- Analysis: The Linear Kernel showed moderate accuracy (0.6374) but failed to predict any positive samples from the minority class, resulting in zero precision, recall, and F1-score. This outcome is expected due to the severe class imbalance in the dataset. This indicates the model's bias towards predicting only the majority class.

### Polynomial Kernel - Air Quality Dataset

- Best Hyperparameters:
  - $C = 0.1$ , Degree = 2,  $c = 0.0$
  - Accuracy = 0.5519
  - Precision = 0.0000
  - Recall = 0.0000

- F1-Score = 0.0000
- Analysis: The Polynomial Kernel achieved an accuracy of 0.5519, but again, it failed to predict any positive samples. The precision, recall, and F1-score remain at 0.0000. This further emphasizes the class imbalance issue, where the model is only predicting the majority class.

### **RBF Kernel - Air Quality Dataset**

- Best Hyperparameters:
  - $C = 0.1$ ,  $\sigma = 0.1$
  - Accuracy = 0.9930
  - Precision = 0.0000
  - Recall = 0.0000
  - F1-Score = 0.0000
- Analysis: The RBF Kernel provided an impressive accuracy of 0.9930, but once again, it failed to detect any positive class samples. The result is consistent with the earlier models, showing that the RBF kernel, while flexible and capable of capturing complex patterns, still suffers from class imbalance.

### **Time Series Kernel - Air Quality Dataset**

- Best Hyperparameters:
  - $C = 0.1$ ,  $\sigma = 1.0$ , Period = 30,  $w_{lin} = 0.5$ ,  $w_{per} = 0.33$ ,  $w_{rbf} = 0.34$
  - Accuracy = 0.9930
  - Precision = 0.0000
  - Recall = 0.0000
  - F1-Score = 0.0000
- Analysis: The Time Series Kernel also achieved an accuracy of 0.9930, but similarly to the other kernels, it predicts only the majority class, resulting in zero precision, recall, and F1-score. Despite its ability to capture temporal patterns, it doesn't address the class imbalance, leading to a bias towards the majority class.

### Comments:

- In all the models tested, the class imbalance led to zero recall, precision, and F1-score for the minority class. This indicates that the models, regardless of the kernel used, were biased towards predicting the majority class.
- The accuracy values are high across all kernels, but this is misleading since the model dominantly predicts the majority class. Accuracy alone is not sufficient for evaluating model performance when class imbalance is present, as it does not reflect the model's ability to detect the minority class.

### Conclusion:

While the Time Series Kernel and RBF Kernel showed the highest accuracy, they still failed to predict any minority class samples due to the class imbalance. Techniques like class weighting or oversampling/undersampling should be applied to improve model predictions for the minority class.

## IMBALANCE HANDLING OUTPUTS

### Imbalance Analysis:

Weather Dataset - Imbalance Ratio: 0.9476

Minority Class: 1773, Majority Class: 1871

Air Quality Dataset - Imbalance Ratio: 0.0222

Minority Class: 35, Majority Class: 1575

### Weather Dataset:

| Method   | Accuracy | Precision | Recall | F1-Score |
|----------|----------|-----------|--------|----------|
| Original | 0.8464   | 0.8583    | 0.8351 | 0.8466   |
| Weighted | 0.8464   | 0.8909    | 0.7946 | 0.8400   |
| SMOTE    | 0.8464   | 0.8583    | 0.8351 | 0.8466   |

### Air Quality Dataset:

| Method   | Accuracy | Precision | Recall | F1-Score |
|----------|----------|-----------|--------|----------|
| Original | nan      | 0.0000    | 0.0000 | 0.0000   |
| Weighted | nan      | 0.0000    | 0.0000 | 0.0000   |
| SMOTE    | nan      | 0.0000    | 0.0000 | 0.0000   |

# DISCUSSION

## WEATHER DATASET

Weather Dataset contains a relatively balanced class distribution with an imbalance ratio of 0.95. This level of imbalance doesn't significantly affect the model's ability to predict both classes. However, handling the imbalance with weighted SVM or SMOTE techniques produced minor shifts in the evaluation metrics.

- Accuracy remained the same across all methods, indicating that the class imbalance wasn't strong enough to drastically impact the overall accuracy.
- Precision increased slightly with the weighted SVM, which applied higher importance to the minority class, leading to a better performance in terms of detecting positive samples.
- Recall decreased for the weighted SVM, which indicates a trade-off when trying to balance precision and recall. The SMOTE method, by generating synthetic data points, allowed the model to detect minority class samples as effectively as the original SVM, with no notable change in F1-Score.

While the imbalance handling techniques (weighted SVM and SMOTE) helped slightly improve precision in the weighted SVM, the SMOTE did not lead to significant improvements over the original model.

## AIR QUALITY DATASET

Air Quality Dataset contains a severe class imbalance with a ratio of 0.0222, where the minority class represents just 2% of the data. This class imbalance severely affects the model's ability to detect the minority class, resulting in zero precision, recall, and F1-score in all methods.

- In the case of the Original SVM, the model fails to predict any positive samples from the minority class, which results in a precision, recall, and F1-score of 0.0000. This is expected since the class imbalance is extreme.
- When weighted SVM was applied, the model also failed to detect any positive samples. The weighted SVM method attempts to penalize the majority class more heavily, but given the highly imbalanced data, this approach still could not mitigate the issue sufficiently.
- The SMOTE technique, which generates synthetic samples for the minority class, also failed to improve the model's ability to detect positive samples. While SMOTE is effective in some situations, it struggled to improve performance in cases of extreme

class imbalance like the Air Quality dataset.